EXTRACTING USER ROLES FROM THE ID TOKEN

In the previous chapter, we hardcoded the list of user roles in the `UserController` class of the Edge Service project because we didn't have them in the ID Token yet. Now that we do, let's refactor the implementation to fetch the roles for the currently authenticated user from the `OidcUser` class, which gives us access to the claims in the ID Token, including the brand-new `roles` claim.

Listing 12.4 Extracting the user roles list from the ID Token via `OidcUser`

```
@RestController
public class UserController {

  @GetMapping("user")
  public Mono<User> getUser(@AuthenticationPrincipal OidcUser oidcUser) {
    var user = new User(
      oidcUser.getPreferredUsername(),
      oidcUser.getGivenName(),
      oidcUser.getFamilyName(),
      oidcUser.getClaimAsStringList("roles")    ❶
    );
    return Mono.just(user);
  }
}
```

❶ Gets the "roles" claim and extracts it as a list of strings

Finally, remember to update the test setup in `UserControllerTests` so that the mock ID Token contains a `roles` claim.

Listing 12.5 Adding `roles` list to the mock ID Token

```
@WebFluxTest(UserController.class)
@Import(SecurityConfig.class)
class UserControllerTests {
```